Michael Ichikawa

AI ENGINEER · AGENTIC SYSTEMS

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SUMMARY

I lead AI integration for a small manufacturer as a fractional engineer, working with the owners on an AI quoting app and an AI assistant. Before that I shipped a production AI agent solo, from an empty repo to listings on the Microsoft Commercial Marketplace and the Teams Store, and built a multi-agent workflow engine from scratch. Earlier I earned an MS in Mathematics, taught college math for eight years, and spent three years in chip design at Intel.

EXPERIENCE

AI Integration Lead (Fractional)

2026 – present

Small manufacturing company (client confidential) · Remote · Contract

- Lead AI integration for the owners and deploy the company's Azure platform and CI/CD pipeline, starting with an AI quoting app and a self-hosted AI assistant
- Build an LLM quoting app that extracts requirements from a customer's whole request (emails, attachments, PDFs and drawings, spreadsheets, photos and scans, and call notes) in seconds and returns product model numbers, each value linked to its source line. The owners say finding that information can take a salesperson half a day
- Separate extraction from decisions. The LLM only extracts and cites its sources verbatim, deterministic rules code built from the company's quoting guide makes every product decision, and the app stops for human review on missing or conflicting information
- Let the owners change quoting rules in natural language, regression-tested against every open and approved quote and 50,000 synthetic requests, with a diff of what would change and approval by a second person
- Ship through CI/CD with staging and live environments per app, rollback, managed identity with no stored secrets, private networking, and 112 automated tests on every pull request

Founder & AI Engineer

2026 – present

Microclaw LLC · Portland, OR · Marketplace listing · Code and evals

- Built Microclaw, one AI agent across Microsoft 365: 87 function-calling tools, a tool-calling loop that chains them until a task is complete, RAG semantic memory, multi-layer failure recovery, and model routing that escalates to a larger model only when a turn needs it
- Redesigned how the agent picks its tools. On 362 blind-written test requests it offered every tool a request needed 98.1 percent of the time, up from 87.0
- On a 142-case agent eval it called the right tools on 134 of 142 cases on average over three runs, with 49 percent fewer prompt tokens than sending every tool
- Took it from first commit to a Marketplace listing in about ten weeks, solo, through Marketplace certification and the Fulfillment and Metering APIs that make it billable, then Teams Store validation after seven review rounds
- Isolated tenants with Postgres row-level security: FORCE policies on every tenant-scoped table, an app role that cannot bypass them, and delegated Entra auth

TypeScript · Azure OpenAI · Microsoft Graph · Azure Postgres (RLS) · Bicep

Career Break · Full-Time Parent, Retraining into Software

2023 – 2026

Portland, OR

- Three years as primary caregiver
- Started in data engineering and ML, then moved to agentic development

Mathematics Faculty

2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught mathematics for eight years at an open-enrollment college: Algebra through Calculus III, Linear Algebra, Differential Equations and introductory Statistics

- Wrote my own lectures and exams, found the topics students kept missing, and reworked how I taught them
- Wrote my own version of an open-source precalculus textbook in LaTeX, reorganized by concept so trig functions came in week 2 instead of week 10

Physical Design Engineer

2008 – 2011

Intel Corporation · Semiconductor R&D

- Owned the layout of whole blocks in the Ivy Bridge (22nm) microprocessor, closing timing, power and design-rule checks
- Kept my blocks passing every check as cell-library updates and upstream circuit changes landed in a large, version-controlled shared design
- Wrote Python and Perl automation for physical verification

PROJECTS

Multi-Agent Workflow Engine

mtichikawa.github.io/workflow-engine

A Python engine, built from scratch on the standard library, that turns a plain-English request into a graph of single-purpose agents. Every workflow is statically validated before it runs, and any step that acts outside the engine must sit behind a human approval gate. A learning loop infers a team's unwritten rules from their corrections, with no fine-tuning: on 150 blind-written messages, routing accuracy rose from 59 to 97 percent. On 320 blind-written patient messages, its triage step sent all 160 medical ones to a person, with no false alarms.

Python · agentic orchestration · MCP (client + server) · static graph validation

Terminal-Bench 3 Task

github.com/mtichikawa/klavis-tb3-submission

An original task for the Terminal-Bench 3 agent benchmark. Two exports of the same orders disagree because of four overlapping defects, and the agent has to recover the right values and work out how many separate causes there are. Claude Code on Claude Opus 5 and Codex on GPT-5.6 failed all six trials. The reference solution scores 1 and doing nothing scores 0, so a zero means the agent failed, not the grader. Every agent got the numbers right and then reported the wrong number of causes.

Open Source Contribution · iTerm2

github.com/gnachman/iTerm2/pull/676

My pull request to the iTerm2 terminal found a gap in how its Lock Layout setting was enforced: lower-level paths for opening tabs, splits and files bypassed the lock. The maintainer audited it and shipped his own fix in master, with public thanks.

Earlier portfolio work

mtichikawa.github.io

Self-directed data engineering and ML projects from 2023 to 2026, including a RAG pipeline on GCP that compares three retrievers (BigQuery Vector Search, FAISS and pgvector) on one eval, streaming and lakehouse pipelines, anomaly detection and forecasting.

TECHNICAL SKILLS

AI-Native Dev	Claude Code · directing AI agents to build, test and debug features, verified by tests and evals
Agents / LLM	agent orchestration · tool-calling agent loops · function calling · tool retrieval (KNN) · RAG (retrieval-augmented generation) · vector search & embeddings (pgvector · FAISS · BigQuery Vector Search) · semantic memory · multi-agent workflows · document extraction with source citations · prompt engineering · human-in-the-loop gating · MCP (client + server) · in-context learning
Reliability & Cost	failure-mode engineering (backoff/retry · context compaction · loop detection · abort ceilings) · model routing with mid-task escalation · input-token reduction · multi-tenant isolation (Postgres RLS)
Evaluation	LLM evaluation · eval design · expected-facts scoring · A/B against baselines · blind held-out test sets · pytest · Vitest · GitHub Actions CI
Built with	Python · TypeScript · Node.js · SQL · Azure OpenAI · Claude API (Anthropic) · Microsoft Graph · PostgreSQL · BigQuery · Delta Lake (medallion) · Spark Structured Streaming · SQLAlchemy · pandas · scikit-learn · HuggingFace · sentence-transformers · Azure (Container Apps · Entra · managed identity · Bicep) · AWS (S3 · Lambda · DynamoDB) · GCP (Vertex AI · Cloud Run) · Docker · FastAPI / REST · Redis

EDUCATION

MS Mathematics — The City College of New York — 2014

BS Mechanical Engineering — University of California, Berkeley — 2008